

Advances in Soft X-Ray Beamline Design: Grating Monochromator Simulations

Theory, Aberration Analysis, and Ray-Tracing Examples

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1. Introduction and Scope

This document summarises the theoretical framework and ray-tracing simulation examples presented by Reininger for soft X-ray (SXR) grating monochromators. Four main instrument types are treated in order of increasing optical sophistication:

1. Spherical Grating Monochromator (SGM)
2. Vertically collimated Plane Grating Monochromator (vertical PGM)
3. Follath collimated PGM (horizontal deflection, fixed focal plane)
4. Focusing Variable Line Spacing PGM (FVLS-PGM)

All simulations use the OASYS/SHADOW ray-tracing environment. Unless otherwise stated, the source parameters correspond to the (then) present APS undulator: a 2 m in-vacuum device with realistic emittance, divergence, and energy spread.

2. Theoretical Foundation: Optical Path Length and Fermat's Principle

2.1. The Optical Path Length Function

The foundation for all grating monochromator designs is the **optical path length function** F for a toroidal grating, expanded in terms of the sagittal coordinate w (across the groove direction) and the meridional coordinate h (along the grooves):

$$F = F_{00} + F_{01} w + \frac{1}{2} F_{20} w^2 + \frac{1}{2} F_{02} h^2 + \frac{1}{2} F_{30} w^3 + \frac{1}{2} F_{21} w h^2 + \dots \quad (1)$$

Fermat's principle requires that, for each diffracted order, $\delta F = 0$ with respect to variations in the ray coordinates. Setting each aberration coefficient to zero yields the design conditions for a perfect (aberration-free) monochromator.

The value of these coefficients are:

$$\frac{1}{d} = k = \frac{\partial n}{\partial w} = \frac{1 + 2b_2w + 3b_3w^2 + \dots}{d_0} \quad (2)$$

$$F_{00} = r + r' \quad (3)$$

$$F_{10} = m \frac{\lambda}{d_0} - (\sin \alpha + \sin \beta) \quad (4)$$

$$F_{20} = \frac{1}{2} \left(\frac{\cos^2 \alpha}{r} - \frac{\cos \alpha}{R} + \frac{\cos^2 \beta}{r'} - \frac{\cos \beta}{R} \right) - \frac{m\lambda b_2}{d_0} \quad (5)$$

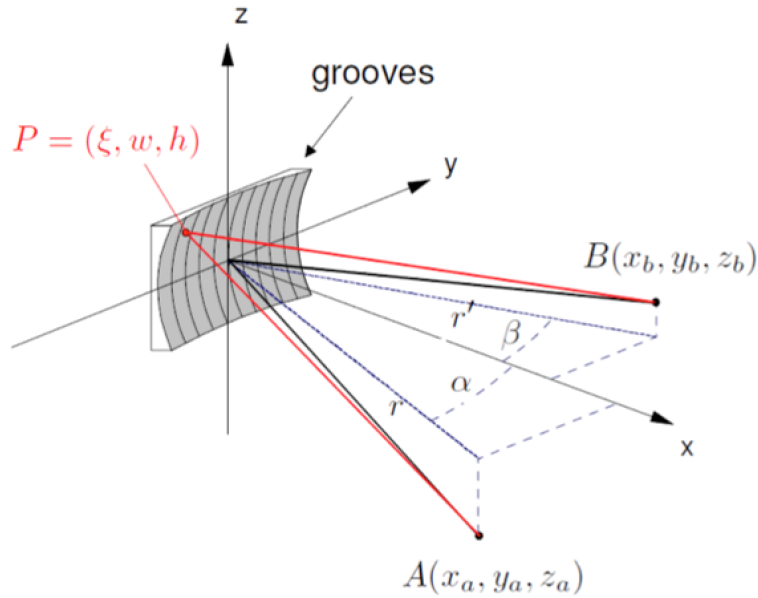
$$F_{02} = \frac{1}{r} - \frac{\cos \alpha}{\rho} + \frac{1}{r'} - \frac{\cos \beta}{\rho} \quad (6)$$

$$F_{12} = \frac{1}{2} \left(\frac{\sin \alpha}{r} \left(\frac{1}{r} - \frac{\cos \alpha}{\rho} \right) + \frac{\sin \beta}{r'} \left(\frac{1}{r'} - \frac{\cos \beta}{\rho} \right) \right) \quad (7)$$

$$F_{30} = \frac{1}{2} \left(\frac{\sin \alpha}{r} \left(\frac{\cos^2 \alpha}{r} - \frac{\cos \alpha}{R} \right) + (\beta r') \right) - \frac{m\lambda b_3}{d_0} \quad (8)$$

Note that for a grating with constant ruling $b_2 = b_3 = 0$.

Figure 1: Geometry of a toroidal grating showing source point S , grating vertex, image point S' , sagittal coordinate h , meridional coordinate w , angles of incidence α and diffraction β , and radii of curvature R (meridional) and ρ (sagittal).



2.2. Meaning of the Coefficients

Term	Name	Effect when non-zero
F_{00}	Central path length	Constant (irrelevant for focusing)
F_{01}	Grating equation term	Selects wavelength; always present
F_{20}	Meridional Focus	Shifts focus along optical axis
F_{02}	Sagittal Focus	Sagittal focus mismatch
F_{30}	Primary coma	Asymmetric blur focal plane

In practice, the key design goals are:

$F_{20} = 0$: meridional focus at the exit slit position:

$$\frac{\cos^2 \alpha}{r} + \frac{\cos^2 \beta}{r'} = \frac{\cos \alpha + \cos \beta}{R} + \frac{2m\lambda b_2}{d_0} \quad (9)$$

$F_{02} = 0$: Sagittal focus at the exit slit position:

$$\frac{1}{r} + \frac{1}{r'} = \frac{\cos \alpha + \cos \beta}{\rho} \quad (10)$$

$F_{30} = 0$: No Primary Coma:

$$\frac{\sin \alpha}{r} \left(\frac{\cos^2 \alpha}{r} - \frac{\cos \alpha}{R} \right) + \frac{\sin \beta}{r'} \left(\frac{\cos^2 \beta}{r'} - \frac{\cos \beta}{R} \right) - \frac{2m\lambda b_3}{d_0} = 0 \quad (11)$$

Note that for the constant ruling grating $b_3 = 0$. On the “Rowland circle” ($r = R \cos \alpha$, $r' = R \cos \beta$), both terms vanish individually, so coma is automatically zero (a good key property of the Rowland circle mounting).

2.3. Grating Equation and Resolving Power

The grating equation in diffraction order m is:

$$m\lambda = d(\sin \alpha - \sin \beta), \quad (12)$$

where d is the groove spacing, α is the angle of incidence, and β is the diffraction angle. Positivs angles are counterclockwise (CCW). In SHADOW notation $m = -1$ corresponds to the first order “internal reflection” ($|\beta| > |\alpha|$).

The theoretical resolving power is determined by the exit-slit-limited resolution:

$$\text{RP} = \frac{\lambda}{\Delta\lambda} = \frac{r' m \cos \beta}{d \cdot \Delta_{\text{slit}}}, \quad (13)$$

where r' is the grating-to-slit distance and Δ_{slit} is the FWHM of the beam at the slit plane.

2.4. Image Size: Sanity Check Formula

A key consistency check in all simulations is the predicted image size at the exit slit, obtained by geometrical demagnification:

$$\Delta_z = 2.35 \sigma_z \frac{\cos \alpha}{\cos \beta} \frac{r'}{r} \quad (\text{vertical/meridional plane}), \quad (14)$$

$$\Delta_x = 2.35 \sigma_x \frac{r'}{r} \quad (\text{horizontal/sagittal plane}), \quad (15)$$

where σ_z , σ_x are the source RMS sizes and r , r' are the source-to-grating and grating-to-image distances. Agreement between the ray-tracing result and Eqs. (14)–(15) confirms that the optical system is properly focused.

3. Spherical Grating Monochromator (SGM)

3.1. Optical Layout

The SGM is the simplest SXR monochromator: a single **concave spherical grating** works simultaneously as the dispersing and focusing element. The geometry is defined by:

- Source-to-grating distance r .
- Grating-to-slit distance r' (movable mechanically).
- Bragg-like incidence angle α and diffraction angle β , both measured from the grating normal.
- Diffraction order $m = -1$.

The defocus condition $F_{20} = 0$ gives the **grating equation for focusing**:

$$\frac{\cos^2 \alpha}{r} + \frac{\cos^2 \beta}{r'} = \frac{\cos \alpha + \cos \beta}{R}, \quad (16)$$

where R is the grating radius of curvature. At each energy, α and β change (coupled through Eq. (12)), so the focal distance r' also changes. Therefore, **the SGM requires a movable exit slit** to track the focus.

SGM design rule: Because $r'(E)$ is energy-dependent, a fixed exit slit position can only be optimal at one energy. At all other energies, coma and defocus degrade the resolution. The exit slit must be translated to follow the focus.

3.2. Ray-Tracing Example at 1000 eV

File: 01_SGM_1000eV.ows

Parameters: $r = 30$ m, $r' \approx 9.9$ m, $\alpha = 87.3^\circ$, $\beta = 89.1^\circ$, $R = 636$ m, $\rho = 9.9$ m, diffraction order $m = -1$.

Results:

- Predicted vertical and horizontal image sizes (sanity check):

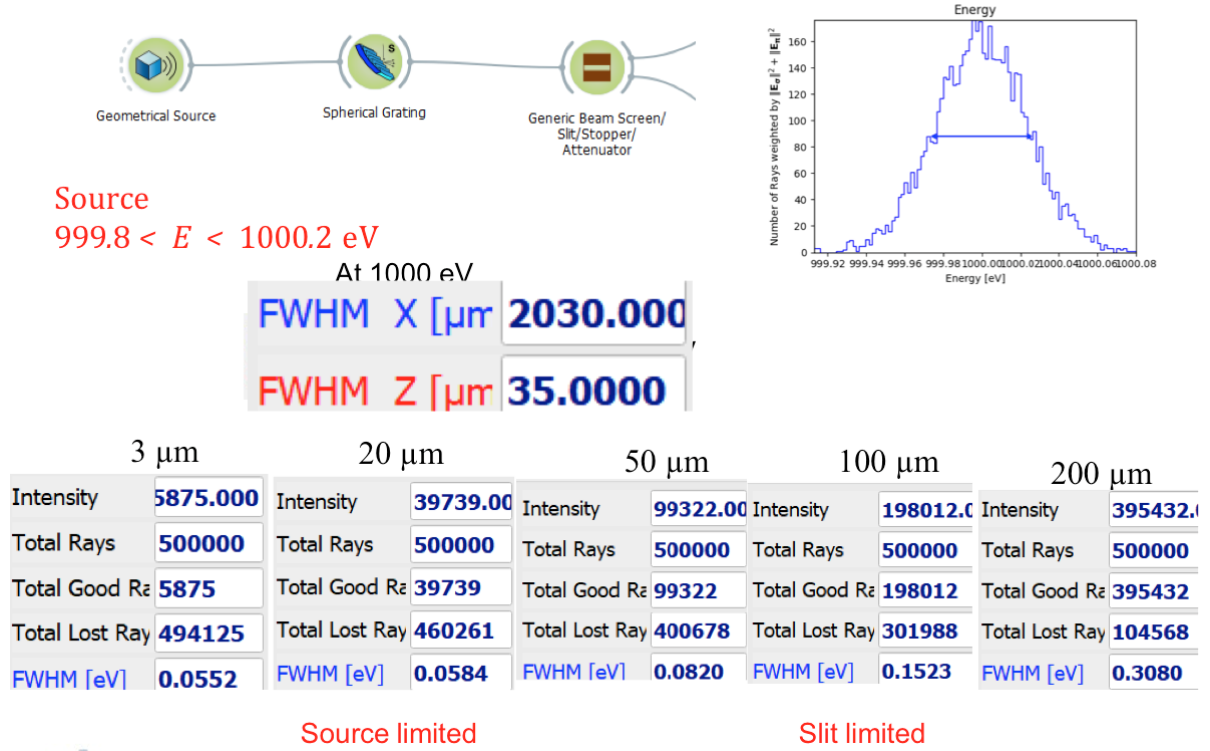
$$\Delta_z = 2.35 \sigma_z (\cos \alpha / \cos \beta) (r'/r) \approx 36 \mu\text{m}.$$

$$\Delta_x \approx 217 \mu\text{m} \text{ (at the position of sagittal focus!).}$$

- Achieved resolving power: $\text{RP} \approx 18\,000$ at 1000 eV with 55 meV FWHM energy resolution.
- Ray-tracing gives $\Delta_z = 35 \mu\text{m}$, $\Delta_x = 2030 \mu\text{m}$.

Slit scan: Using source in the photon energy interval $999.8 < E < 1000.2 \text{ eV}$, the resolution is obtained from the FWHM of the intensity-energy histogram. Using a slit of $3 \mu\text{m}$, the resulting $\Delta E = 0.552 \text{ eV}$ ($\text{RP} = 18116$). It is verified that roughly constant resolution is obtained slit openings $\lesssim 20 \mu\text{m}$ (*source limited*). For larger openings ($\gtrsim 50 \mu\text{m}$) the resolution is *slit limited*, degrading linearly with slit width.

Figure 2: SHADOW ray-tracing results for SGM at 1000 eV , including spot dimensions at the exit slit plane, energy resolution histogram, and energy resolution vs. slit opening, showing source-limited and slit-limited regimes with the transition near $20 - 50 \mu\text{m}$.



3.3. Ray-Tracing Example at 1600 eV

File: 02_SGM_1600eV.ows

Parameters: $\alpha = 87.63^\circ$, $\beta = -88.77^\circ$. Source: $\Sigma_x = 280 \mu\text{m}$, $\Sigma_z = 13.7 \mu\text{m}$, $\Sigma'_x = 18.4 \mu\text{rad}$, $\Sigma'_z = 14.2 \mu\text{rad}$.

Key finding — focus shift with energy:

The sanity-check formula predicts $\Delta_z \approx 20 \mu\text{m}$ at the nominal slit position, but the ray-tracing footprint shows the beam is *not at focus* at 1600 eV if the exit slit remains

at the 1000 eV position (the vertical size is $55.6 \mu\text{m}$ FWHM). Moving the exit slit to $r' = 11\,046 \text{ mm}$ (satisfying $F_{20}(1600 \text{ eV}) = 0$) restores focus and gives $\Delta_z \approx 22 \mu\text{m}$, consistent with the sanity check.

Conclusion: This example confirms that the SGM requires slit translation with energy. The exit slit movement *corrects defocus but does not correct coma*.

4. Plane Grating Monochromator (PGM)

4.1. Key Properties of a Plane Grating

A plane grating (without variable line spacing) **does not focus**. Its role is purely dispersive. The focusing is delegated entirely to curved mirrors upstream and downstream of the grating. The important consequence is a **fixed focal plane**: the image distance is independent of the grating rotation.

The **grating magnification** factor is c^{-1} , with c defined using the Rowland circle condition:

$$c = \frac{r'}{r} = \frac{R \cos \beta}{R \cos \alpha} = \frac{\cos \beta}{\cos \alpha}, \quad (17)$$

and the grating equation becomes:

$$\left(\frac{m\lambda}{d} - \sin \beta \right)^2 = 1 - \frac{1 - \sin^2 \beta}{c^2}. \quad (18)$$

The parameter c controls the angular compression/expansion between the collimated beam and the diffracted beam, and thus the grating magnification c^{-1} can be tuned by rotating the plane mirror.

4.2. Collimated PGM: Vertical Deflection Layout

File: 03_CollimatedPGM1000eVc2_Vertical.ows

Layout (distances from source): Toroidal mirror at 28 m $\rightarrow$ plane mirror + plane grating at 2 m downstream from the Toroidal mirror. Cylindrical mirror at 1 m downstream from the grating. Image plane 9 m further downstream.

Optical roles:

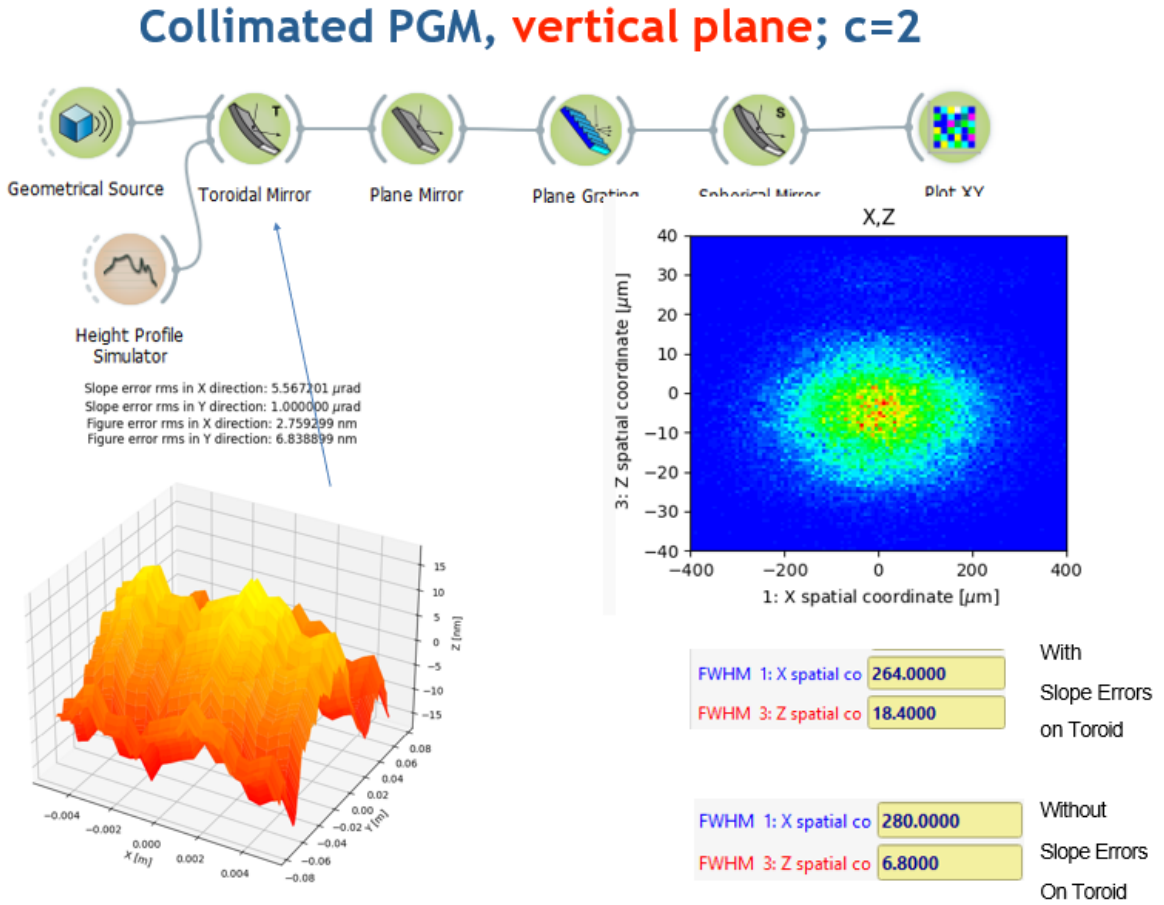
- **Toroidal mirror:** collimates the beam in the meridional (vertical) plane; focuses sagittally (horizontally) at the exit slit.
- **Plane mirror + plane grating:** deflect in the vertical plane; the plane mirror rotation controls c and hence the wavelength.
- **Cylindrical mirror:** focuses the collimated, dispersed beam onto the *fixed* exit slit.

Parameters at 1000 eV, $c = 2$: $\alpha = 88.5^\circ$, $\beta = 87.1^\circ$; plane mirror at 87.8° incidence.

Aberrations:

- **Coma** is present in the vertical plane because the collimated beam does not enter the grating exactly at normal incidence (residual F_{30}).
- Image size at slit: $\sim 5.8 \mu\text{m}$ (theoretical), $\sim 6.8 \mu\text{m}$ (ray-tracing, evidencing coma).
- **Slope errors:** It is shown that the slope errors on the toroidal mirror broaden the image significantly at the image plane (passing from $6.8 \mu\text{m}$ to $18.4 \mu\text{m}$). Therefore, the slope errors on the upstream toroid are the dominant resolution-limiting factor in this configuration.

Figure 3: Effect of slope errors on the toroidal mirror in the vertical PGM. The image on the slit plane with realistic r.m.s. slope errors is $\sim 18.4 \mu\text{m}$ (ideal, no slope errors: $\sim 5.8 \mu\text{m}$). This demonstrates the degradation of the resolution due to the slope errors that affect the collimation in the diffraction plane.



4.3. Follath Collimated PGM: Horizontal Deflection

The **Follath PGM** (R. Follath and F. Senf, NIM A 390, 388, 1997) uses the toroidal mirror, such as the **horizontal** sagittal focusing is in the grating vertical (diffraction) plane. The key advantage is that, in the sagittal plane, slope errors on the toroidal mirror are much less critical because the sagittal focusing is less sensitive to meridional slope

errors. This is expressed by the forgiveness factor¹

File: 04_CollimatedPGM1000eVc2.ows

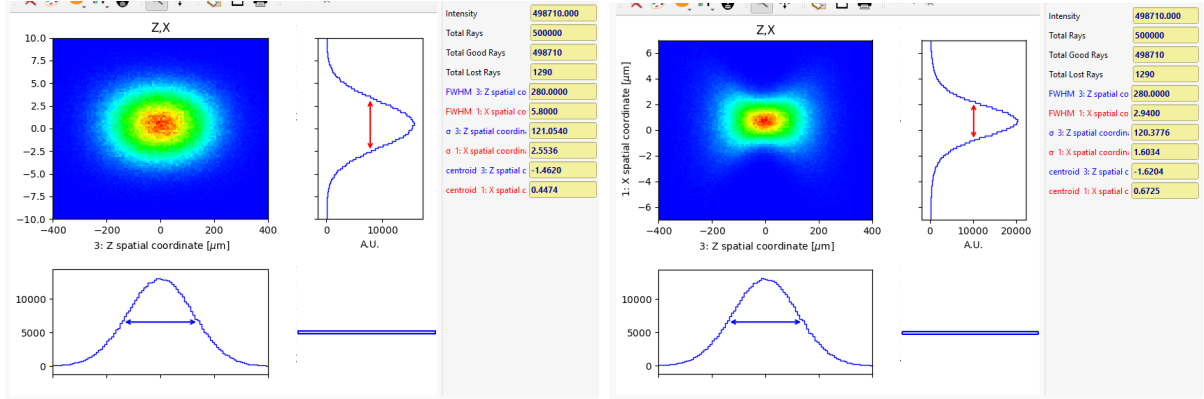
Same layout as above, but mirror–grating system rotates in the horizontal plane.

Key results at $c = 2$, 1000 eV:

- **No coma:** the horizontal geometry naturally satisfies $F_{30} \approx 0$ due to the relative rotation between the plane mirror and the grating in the xz -plane.
- Image size: $\sim 5.8 \mu\text{m}$ (same as vertical case without aberrations, but now robust against mirror slope errors).
- The “**forgiveness factor**” $\times \theta$: slope errors on the toroidal mirror have a much smaller effect in the horizontal geometry.
- Simulation with and without slope errors confirms that the Follath layout is significantly more tolerant to mirror figure errors.

Effect of increasing c : At $c = 5$ (file 05_CollimatedPGM1000eVc5.ows), $\alpha = 89.5^\circ$, $\beta = 87.4^\circ$. The **astigmatic coma** grows, scaling as $\sim 5.8/(c/2) = 2.32 \mu\text{m}$ for ideal optics. With slope errors, the image quality degrades further.

Figure 4: Left: Follath horizontal PGM ($c = 2$, 1000 eV). Observe: no coma, practically no effect of slope error. Right: Follath horizontal PGM ($c = 5$, 1000 eV). Observe: Astigmatic coma (term F_{12}), small effect of slope error.



5. Focusing Variable Line Spacing PGM (FVLS-PGM)

5.1. Principle

The **FVLS-PGM** (R. Reininger and A. R. B. de Castro, NIM A 538, 760, 2005) addresses the two main limitations of the collimated PGM:

¹In the meridional plane, the slope errors σ increase the focal size by a factor $2q\sigma$. In the sagittal direction, the same slope error increases the size by a factor $2q\sigma \sin \theta$, where θ is the grazing angle of incidence. Since $\theta \ll 1$ for X-ray mirrors, the sagittal contribution is reduced by the *forgiveness factor* $\sin \theta \ll 1$.

1. **Heat-load-induced deformations** of the plane mirror degrade the resolution by introducing a virtual source shift.
2. **Coma** cannot be simultaneously cancelled at all energies with a fixed line-spacing grating.

The solution is to use a **variable line spacing** (VLS) grating, where the groove density varies along the grating surface as:

$$k = \frac{1 + 2b_2 w + 3b_3 w^2 + \dots}{d_0}, \quad (19)$$

with $1/d_0$ the central line density. The coefficients b_2 and b_3 are chosen to:

- Choose the grating magnification $c^{-1} = (\cos \beta / \cos \alpha)^{-1}$.
- Choose λ_0 for which b_2 will be zero.
- Solve the grating equation 12 for β_0 .
- b_2 : Solve the defocus equation $F_{20} = 0$.
- b_3 : Cancel coma $F_{30} = 0$ at one selected wavelength.

FVLS design parameters (example at central energy 600 eV, $c = 2$):

$$b_2 = 1.44 \times 10^{-4} \text{ mm}^{-1},$$

$$b_3 = 1.3 \times 10^{-8} \text{ mm}^{-2}.$$

Coma is zeroed at 600 eV; solving coma at one wavelength is sufficient because the residual coma at other energies remains small.

5.2. Optical Layout

File: 06_FVLSPGM1000eVc2M2WithWithout.ows

Distances: Source $\rightarrow$ heat absorber (placed artificially at 0 m) $\rightarrow$ plane mirror (29.7 m) $\rightarrow$ VLS grating (+0.3 m) $\rightarrow$ slit plane (+10 m) .

Grating designed for $c = 2$ at 600 eV.

Parameters at 1000 eV: $\alpha = 88.53^\circ$, $\beta = 87.05^\circ$ (same $c = 2$ condition).

Key feature: There is no upstream collimating mirror. The VLS grating itself provides the meridional focusing at the exit slit, and the horizontal focus is provided after the slit by a separate refocusing mirror.

5.3. Heat Load and Correction

The crucial advantage of the FVLS-PGM emerges when heat-load deformations of the plane mirror are considered:

Heat load simulation: The plane mirror absorbs ~ 35 W of total power. The resulting thermo-mechanical deformation produces a maximum surface height deviation of $\sim 0.40 \mu\text{m}$ and slope errors up to $\sim 15 \mu\text{rad}$ peak. The effective radius of curvature

induced is $R \approx -30$ km (very weakly concave). This shifts the virtual source position to ~ 28.52 m.

Without correction: the deformed mirror creates a defocused, asymmetric image at the slit — resolving power degrades substantially.

Correction strategy: Because the FVLS grating corrects defocus at all energies by adjusting the angle of incidence, a *small angular correction* to α and β is sufficient to restore the focus:

$$\alpha : 88.52843^\circ \rightarrow 88.53789^\circ, \quad \beta : 87.05389^\circ \rightarrow 87.05860^\circ.$$

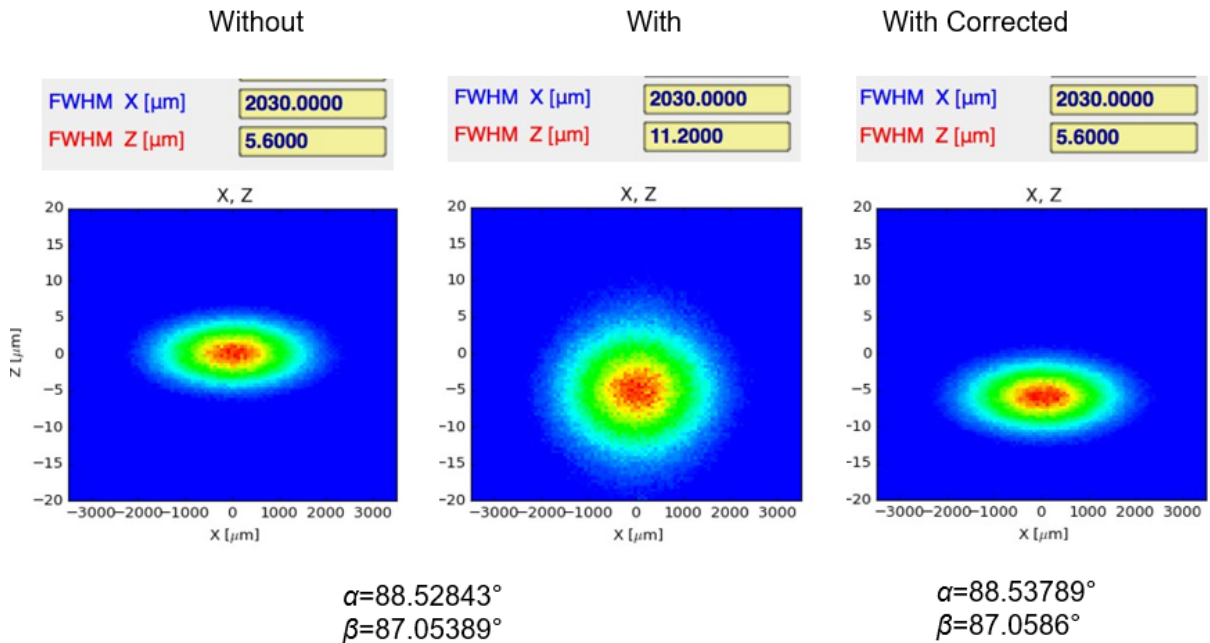
This is equivalent to accounting for the shifted virtual source position in the VLS design equations.

Result: After angular correction, the ray-tracing shows full recovery of the nominal slit image — the FVLS-PGM **self-corrects for heat-load deformations** with only a minor adjustment of the mirror-grating angles.

Reference: R. Reininger (2010).

Limitation of the FVLS-PGM: Because c is fixed by the VLS coefficients (b_2), the grating magnification *cannot* be changed during operation. The Follath collimated PGM retains the freedom to vary c (and hence the resolving power vs. flux trade-off), but at the cost of reduced heat-load tolerance.

Figure 5: FVLS-PGM ray-tracing at the exit slit under three conditions: (a) no heat load (reference); (b) with heat load, uncorrected (defocused, RP degraded); (c) with heat load, corrected by angular adjustment ($\Delta\alpha = +0.00946^\circ$, $\Delta\beta = +0.00471^\circ$) — full RP recovery.



6. Comparison of Monochromator Types

Property	SGM	Vertical PGM	Follath PGM	FVLS-PGM
Exit slit position	Movable (tracks focus)	Fixed	Fixed	Fixed
Coma	Present	Present	Eliminated (hor. geom.)	Zeroed at one λ
Slope error sensitivity	Moderate	High (toroid)	Low (forgiveness factor)	Low
Heat load correction	Slit movement	Slit movement	Slit movement	Angle correction
Variable magnification c	N/A	Yes	Yes	No (fixed by VLS)
Focusing element	Grating only	Toroid cylinder	+ Toroid cylinder	+ VLS grating

7. Summary and Conclusions

A unified theoretical framework — the optical path length expansion of Eq. (1) and Fermat’s principle — underpins all grating monochromator designs for soft X-rays. The main conclusions are:

1. **The SGM** is conceptually simple but operationally demanding: the exit slit must track the focus with energy, and coma cannot be corrected independently.
2. **The collimated PGM** decouples focusing from dispersion. The *vertical* layout is coma-affected and sensitive to toroidal mirror slope errors. The *Follath* layout eliminates coma and is far more tolerant to slope errors (the “forgiveness factor”).
3. **The FVLS-PGM** is the most robust design: VLS coefficients simultaneously cancel defocus at all energies and coma at the design energy, and the grating angles can be slightly adjusted to compensate heat-load deformations of the plane mirror without moving the exit slit.
4. **OASYS/SHADOW** provides the numerical environment to implement all these designs and verify them against the analytical sanity checks (image size, resolving power). The formulas for all monochromator types discussed here are included in the OASYS widget library.

Glossary of Symbols

Symbol	Definition
F	Optical path length function
F_{ij}	Coefficients of the OPL expansion (i : meridional order, j : sagittal order)
w	Coordinate along the groove direction (sagittal)
ℓ	Coordinate across the grooves (meridional)
α	Angle of incidence on the grating (from grating normal)
β	Angle of diffraction
m	Diffraction order
d	Groove spacing (at grating centre)
λ	Wavelength
r	Source-to-grating distance
r'	Grating-to-image (exit slit) distance
R	Grating radius of curvature (Meridional)
ρ	Grating radius of curvature (Sagittal)
c	Inverse of Magnification factor ($c = \cos \beta / \cos \alpha$)
RP	Resolving power ($RP = \lambda / \Delta \lambda = -E / \Delta E$)
Σ_x, Σ_z	Source size (horizontal, vertical) RMS
Σ'_x, Σ'_z	Source divergence (horizontal, vertical) RMS
b_2, b_3	VLS grating coefficients (line density gradient, curvature)
$1/d_0$	Central groove density of VLS grating

References

1. R. Follath and F. Senf, *New plane grating monochromators for third generation synchrotron radiation light sources*, Nucl. Instrum. Methods Phys. Res. A **390**, 388 (1997). doi:10.1016/S0168-9002(97)00401-4
2. R. Reininger and A. R. B. de Castro, *The Focusing VLS-PGM: A new monochromator for soft X-rays*, Nucl. Instrum. Methods Phys. Res. A **538**, 760 (2005). doi:10.1016/j.nima.2004.09.007
3. R. Reininger, *Correcting Heat Load Induced Deformations on a XUV Beamline*, Proc. of ACTOP2008, Second Workshop on X-ray and XUV Optics, Trieste (Italy) (2008).
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